This assignment will not be graded and is only for practice.

1) Choose the set of correct options. *1 point*

- [ ] Any subset of any linearly independent set of vectors is linearly independent.
- [ ] Any subset of any linearly dependent set of vectors is linearly dependent.
- [ ] Any subset of any linearly dependent set of vectors is linearly independent.
- [ ] Any set of 3 vectors in $\mathbb{R}^3$ is linearly independent.
- [ ] Any set of 4 vectors in $\mathbb{R}^3$ is linearly dependent.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Any subset of any linearly independent set of vectors is linearly independent.
Any set of 4 vectors in $\mathbb{R}^3$ is linearly dependent.

Recall the definition of linearly dependent and independent:
Statement 1) Definition of linear dependence: A set of vectors $\{v_1, v_2, \ldots v_n\}$ in V is said to be linearly dependent if there exist scalars $a_1, a_2, \ldots a_n \in \mathbb{R}$, not all zero such that $a_1v_1 + a_2v_2 + \ldots + a_nv_n = 0$.
Statement 2) Definition of linear independence: A set of vectors $\{v_1, v_2, \ldots v_n\}$ in V is said to be linearly independent if $a_1v_1 + a_2v_2 + \ldots + a_nv_n = 0$ implies that $a_i = 0$ for $i = 1, 2, \ldots n$.
Statement 3) If S is a set containing zero vector, then the set S is a linearly dependent set.
Statement 4) Let S be a subset of a vector space V. If an element of S is a scalar multiple of another element from the set S, then S is a linearly dependent set.
Statement 5) If S is linearly dependent, then there exists $v \in S$, such that v can be written as a linear combination of other vectors in S and vice versa.

Discussion on Option 1: Let $S = \{v_1, v_2, \ldots, v_n\}$ be a linearly independent set of vectors and let S' be a subset of S.
Claim: S' is linearly independent.

2) **What we have to assume for S' if we want to use the method of contradiction?** *1 point*

- ( ) Let us assume that S' is linearly dependent.

◯ Let us assume that $S^{'}$ is linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Let us assume that $S^{'}$ is linearly dependent.

If $S^{'}$ is linearly dependent, then there exists $v \in S^{'}$, such that v can be written as a linear combination of other vectors in $S^{'}$.

3) Which of the above statements we are using here?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

1 point

$v \in S^{'}$ implies $v \in S$, as $S^{'} \subseteq S$. Those vectors of whose linear combination is giving v are also in S due to the same argument.

4) Again from statement 5, can we conclude that S is linearly dependent? ***1 point***

◯ Yes.

◯ No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Yes.

But S was given to be a linearly independent set, hence we arrive at a contradiction. So our claim is proved.
Another method: Let $S^{'} = \{v_{i_1}, v_{i_2}, \ldots, v_{i_k}\} \subseteq S$. Consider the equation

$$a_1 v_{i_1} + a_2 v_{i_2} + \ldots + a_k v_{i_k} = 0$$

5) Can we conclude each $a_i = 0$ from here? ***1 point***

- Yes.
- No.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Yes.

As all the v_{i_j} are also in S, the given equation is a linear combination of vectors from S. Now as S is linear independent, $a_i = 0$ for each $i \in \{1, 2, \ldots, k\}$. Hence $S^{'}$ is linearly independent.
· **Now try Question 7 and 8 of activity 3.3**

Discussion on Option 2: Let us discuss an example. Let $S = \{(1,0,0), (0,1,0), (1,1,0)\}$ be a subset of R^3.

6) Is S linearly dependent? ***1 point***

- Yes.
- No.

No, the answer is incorrect.
Score: 0

Feedback:

Hint: (1, 1, 0) = (1, 0, 0) + (0, 1, 0)
Accepted Answers:

Yes.

7) Can you find a linearly independent subset of S? **1 point**

- Yes.
- No.

No, the answer is incorrect.
Score: 0

Feedback:

Hint: Check that any proper subset of S of cardinality 1 or 2 is linearly independent subset of S.

Accepted Answers:

Yes.

Consider $S^{'} = \{(1,0,0),(0,1,0)\}$.

8) Is $S^{'} \subseteq S$? **1 point**

- Yes.
- No.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Yes.

9) Is $S^{'}$ linearly independent? **1 point**

- Yes.
- No.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Yes.

Hence Option 2 is not correct.

Discussion on Option 3: Here also we construct an example. Let $S = \{(1,0,0),(2,0,0),(3,0,0)\}$.

10) Is S linearly dependent? ***1 point***

- Yes.
- No.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Yes.

Consider $S' = \{(1,0,0),(2,0,0)\}$.

11) Is $S' \subseteq S$? ***1 point***

- Yes.
- No.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Yes.

12) Is $S^{'}$ linearly independent? ***1 point***

- Yes.
- No.

No, the answer is incorrect.
Score: 0

Accepted Answers:

No.

Hence Option 3 is not correct.

Discussion on Option 4: Consider the same set $S = \{(1,0,0),(2,0,0),(3,0,0)\}$ that we have discussed previously.

13) Is S linearly independent? ***1 point***

- Yes.
- No.

No, the answer is incorrect.
Score: 0

Accepted Answers:

No.

14) What can we conclude about Option 4 from this? **1 point**

- Option 4 is correct.
- Option 4 is not correct.
- Nothing can be concluded about Option 4 from here.

No, the answer is incorrect.
Score: 0

Accepted Answers:

Option 4 is not correct.

· **Now try Question 5 and 8 of activity 3.4**

Discussion on Option 5: Let $S = \{v_1, v_2, v_3, v_4\}$ be a set of 4 vectors in $\mathbb{R}^3$. Let $v_1 = (x_1, y_1, z_1)$, $v_2 = (x_2, y_2, z_2)$, $v_3 = (x_3, y_3, z_3)$, $v_4 = (x_4, y_4, z_4)$. Consider the following equation,

$$a_1v_1 + a_2v_2 + a_3v_3 + a_4v_4 = 0$$

i.e.,

$$a_1(x_1, y_1, z_1) + a_2(x_2, y_2, z_2) + a_3(x_3, y_3, z_3) + a_4(x_4, y_4, z_4) = (0, 0, 0)$$

i.e.,

$$\begin{aligned} a_1x_1 + a_2x_2 + a_3x_3 + a_4x_4 &= 0 \\ a_1y_1 + a_2y_2 + a_3y_3 + a_4y_4 &= 0 \\ a_1z_1 + a_2z_2 + a_3z_3 + a_4z_4 &= 0 \end{aligned}$$

The matrix representation of this system of linear equations is given by

$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 \\ y_1 & y_2 & y_3 & y_4 \\ z_1 & z_2 & z_3 & z_4 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

We want to find the solution for the vector $\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}$ for this given homogeneous system of linear equations.

15) What can we say about the solutions of this system of linear equations? ***1 point***

- [ ] The solution is unique.
- [ ] There are infinitely many solutions.
- [ ] There must exist a non-zero solution.
- [ ] 0 is the only solution.

No, the answer is incorrect.

Score: 0

Feedback:

Hint: Try to compute the maximum number of dependent variables you can get from the row reduced echelon form of the above matrix.

Accepted Answers:

There are infinitely many solutions.
There must exist a non-zero solution.

16) Is S linearly independent? ***1 point***

- ( ) Yes.
- ( ) No.

No, the answer is incorrect.

Score: 0

Accepted Answers:

No.

Hence Option 5 is correct.

·**This will help you to solve Question number 4 of the graded assignment.**

Your score is: 0/16